

GARRETT COMES

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SUMMARY

Aerospace engineer (Georgia Tech, propulsion concentration) working at the intersection of spacecraft fluid/propulsion hardware and scientific software. Four summers at Lockheed Martin Space on flight programs including Dragonfly and Mars Sample Return. Builds production-quality simulation tools — GPU CFD solvers, superconducting-coil multiphysics, orbital analysis — in Rust, Python, and CUDA, with independent research on probabilistic/thermodynamic computing hardware. Strong background in engineering optimization and agentic AI tooling.

EDUCATION

Georgia Institute of Technology

Jul 2022 – Aug 2026 (expected)

B.S. Aerospace Engineering, Concentration in Propulsion

Atlanta, GA

– Coursework: Jet & Rocket Propulsion, Computational Fluid Dynamics, Aerothermodynamics, Finite Element Analysis, Control System Design, Aircraft Flight Dynamics, Aerodynamics.

PROFESSIONAL EXPERIENCE

Lockheed Martin Space — Propulsion Engineering Intern, Civil & Commercial Space

Summers 2022 – 2025

Flight programs incl. Dragonfly and Mars Sample Return

Waterton Campus, Denver, CO

- Designed a piston-tank propellant gauging system from a blank sheet: internal hermetically sealed magnets sensed by radiation-hardened Hall-effect sensors; owned magnetic circuit sizing, sensor selection, magnetic simulation, and integration concept, then correlated analysis with bench testing.
- Built propellant slosh CFD simulations (Ansys) of spacecraft tanks and supported design of a slosh characterization test rig used to correlate model predictions against test data.
- Contributed to a green-propellant trade study evaluating hydrazine-replacement monopropellants across performance, materials compatibility, and handling/safety criteria.
- Designed and analyzed propulsion hardware and fluid-system components — valves, interfaces, and test support equipment — including Creo CAD models, layouts, and drawings for propulsion assemblies.
- Supported propulsion hardware test campaigns, requirements verification, anomaly investigation, and cross-functional design reviews with systems, manufacturing, and test teams.

PUBLICATIONS

- **SCRAP: Contactless Detumbling and Capture of Orbital Debris via HTS Coil** — extended abstract in preparation, AIAA SciTech 2027; co-authors from Lockheed Martin, Astranis, and Boeing. [draft] (2026)

INDEPENDENT RESEARCH

Thermodynamic-Computing Sampling Research | JAX, THRML, PyTorch, flash-attn

2026

- Hybridized a pretrained masked-diffusion language model with block-Gibbs joint sampling over a factor graph on Extropic's THRML thermodynamic-hardware simulator; joint factor-graph decoding beat independent argmax by 7–15 pp on a held-out test split, with hard-equality factor maps dominating.
- Ran fail-fast probes on the “no classical escape” hypothesis and retired it: warm-started AIS/SMC crossed every tested energy barrier, closing the per-sample-accuracy axis for TSU advantage claims.
- Quantified a sampling coverage frontier across 59 iid $q=20$ Potts landscapes (protein-design-style coevolutionary models); derived the end-to-end hardware energy ratio required for crossover at digital budgets and labeled it explicitly as unmeasured.
- Subjected all headline claims to adversarial review gates that scoped an initial $10^6 \times$ energy claim into a precise conditional one; final pipeline reproduces bit-for-bit with 9/9 regression tests passing.

ENGINEERING PROJECTS

SCRAP — Orbital Debris Detumbling & Capture (Senior Design) | Python, FEniCSx, Rust, Gmsh

2025 – 2026

- Developed a coupled electromagnetic–thermal quench simulation of the YBCO HTS pancake coil (H-formulation, E–J power-law resistivity, per-turn tape meshing, radiation/cold-finger BCs, quench detection, adaptive time-stepping) to establish safe operating current margins.
- Trained DDPG reinforcement-learning agents for eddy-current detumbling of uncooperative tumbling targets.

spacedb — Space-Object Database & Servicing Route Planner | Rust, egui, wgpu

2025 – 2026

- Desktop GUI + CLI over the full public space-object catalog (Space-Track, GCAT, ESA DISCOS): SGP4 and high-fidelity propagation, collision screening, decay prediction, debris value scoring, and multi-target servicing-route optimization; wgpu-rendered interactive 3D Earth/orbit scene.

Valurile — GPU Lattice-Boltzmann CFD Solver | Rust, CUDA

2024 – present

- Multi-crate LBM workspace: Esoteric Pull in-place streaming ($\sim 2 \times$ memory-bandwidth efficiency), static mesh refinement, Smagorinsky–Lilly LES, SRT/TRT/MRT collision operators, airfoil/cylinder obstacles with drag/lift histories, and interactive egui/GPU visualization.

fem_rust_ae — FEM Solver in Rust (live browser demo) | Rust, WASM, Bevy

2025

- From-scratch structural FEM (shell model of a cylinder under axial load) with sparse solve via faer, interactive Bevy/egui 3D UI, mesh-refinement convergence studies, and a WebAssembly build running live in the browser.
- ai-cfd — GPU-Resident Flow Solver for ML Dataset Generation** | *Python, Triton* 2026
 - Staggered-grid incompressible cylinder-flow solver written entirely in Triton kernels (momentum, projection, rendering); adaptive timestep, Jacobi/CG/BiCGStab pressure options, per-step physics diagnostics, and train/val/test sweep generation for AI-CFD surrogate experiments.
- repx — eBPF Reproducible-Build Attestation** | *Rust, Aya eBPF* 2026
 - Kernel-level tracing of build process/file events canonicalized into Merkle-rooted attestations with content-addressed dataflow-DAG provenance; verification reruns report divergent tree nodes, not just pass/fail.
- any-nix-bootstrap — Trust-Minimized Software Bootstrap** | *POSIX sh, live-bootstrap, Guix, Nix* 2026
 - Bootstrap chain from an 813-byte auditable seed through GCC 15/musl/Guile, GNU Guix, and Nix to any nixpkgs package, fully offline from one hash-verified source manifest; stage 1 executed end-to-end on real hardware (314 hash-pinned distfiles).
- Eddy-Current Braking System** | *Rust no_std, Arduino Due, NGSolve* 2025
 - Embedded controls + magnetic simulation: no_std firmware with AS5048A encoder feedback, H-bridge PWM, feedforward/PI speed control, UART telemetry to a live egui tuning dashboard, and Python/NGSolve eddy-current braking studies.

SKILLS

Languages: Python, Rust, C++, MATLAB/Simulink, CUDA/Triton, Wolfram, LaTeX, shell
Simulation & CAE: Ansys (Fluent/Mechanical), FEniCSx, NGSolve, Gmsh, SolidWorks (CSWA), Creo, Fusion 360, Inventor
Scientific computing: JAX, PyTorch, NumPy/SciPy, wgpu/OpenGL, MPI/PETSc, probabilistic inference (Gibbs/AIS/SMC), engineering optimization
Aerospace: space propulsion & fluid systems, valves, propellant management, orbital mechanics (SGP4/high-fidelity propagation), CFD & slosh analysis, magnetic simulation & test
Systems & tooling: Linux/NixOS, Git, eBPF, embedded (no_std Rust), agentic AI development (Claude Code, MCP servers, custom agent frameworks)

LEADERSHIP & ACTIVITIES

Yellow Jacket Space Program <i>Propulsion — pressurant isolation valve design for a collegiate liquid-propellant space-shot vehicle</i>	2023 – present <i>Georgia Tech</i>
Apex Robotics (FRC/VEX) <i>President & Lead Software Engineer — competition robots, outreach, fundraising, team coordination</i>	2017 – 2022 <i>Mesa, AZ</i>
Stellar Xplorers <i>Team Captain — led team to one of ten national-finals spots (Houston)</i>	2021 – 2022 <i>Mesa, AZ</i>

CERTIFICATIONS & AWARDS

Certifications: SolidWorks CSWA · Autodesk Inventor · Autodesk Fusion 360 · Wolfram Language Level 1
Awards: CTE Presidential Scholar — National Semifinalist (2022)